

# **Machine Learning IMP Bank for Mid-Term Exam By MCA Scholar's Group**

MCA 1<sup>st</sup> Sem (Pattern 2024)

## **Unit 1: Introduction to ML**

1. Define **Machine Learning** and explain its significance.
2. Differentiate between **Machine Learning and Statistical Learning** with examples.
3. Explain different **Types of Machine Learning** (Supervised, Unsupervised, and Reinforcement Learning) with examples.
4. Describe the **Machine Learning Workflow** and explain each step in detail.
5. What are **Features, Labels, Datasets** in ML? Explain with examples.
6. Define **Overfitting and Underfitting** in ML. How can they be prevented?
7. Explain the **Bias-Variance Tradeoff** with examples.

## **Unit 2: Supervised Learning**

1. What is **Supervised Learning**? Explain its key characteristics.
2. Explain **Simple Linear Regression and Multiple Linear Regression** with mathematical formulas and examples.
3. Compare different **Classification Algorithms** (Logistic Regression, Decision Trees, k-NN, SVM, Naïve Bayes) and their applications.
4. Discuss **Evaluation Metrics** like **Accuracy, Precision, Recall, and F1-Score**. Why are they important?
5. Explain the concept of **Train/Test Split and Cross-Validation** in model evaluation.
6. What is a **Confusion Matrix**? Explain how it is used in model evaluation with an example.

## **Unit 3: Unsupervised Learning**

1. Define **Unsupervised Learning** and explain its purpose with real-world applications.
2. Compare different **Clustering Algorithms** (K-Means, Hierarchical Clustering, DBSCAN) and explain their working principles.
3. Would you like explanations or Python implementations for any specific questions?